Practice Problems
Calculus Physics II

Name: Key **Date:** _____

Chapter 46 Magnetic Forces

1) Moving electric charges will interact with

A) only an electric field.
B) an electric field or a magnetic field.
C) only a magnetic field.
D) none of these

2) The force on an electron moving in a magnetic field will be the largest when its direction is

A) the same as the magnetic field direction.
B) perpendicular to the magnetic field direction.
C) at an angle other than 90 degrees to the magnetic field direction.
D) exactly opposite to the magnetic field direction.
E) none of these

$|\vec{F}_m| = |q||\vec{v}||\vec{B}| \sin(\phi)$

max at 90°

3) Which field can increase a moving electron's speed?

A) only a magnetic field
B) either an electric or magnetic field
C) only an electric field
D) none of these

$\vec{E}$ - field only changes direction of object with net charge

4) Which force field can accelerate an electron, but never change its speed?

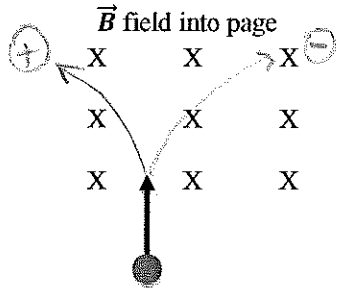
A) electric field
B) magnetic field
C) both of these
D) none of these

Circular motion

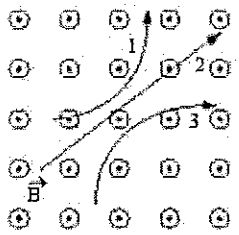
5) A charged sphere enters a magnetic field as shown, which direction will the magnetic force act when the sphere first enters the field

RHR #1

	Positive charge sphere	Negative charged sphere
A)	right	Into the page
B)	Into page	Out of page
C)	Right	left
D)	left	right
E)	Out of page	Into page

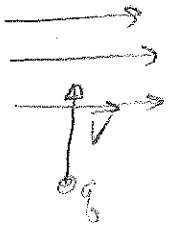


6) Three particles travel through a region of space where the magnetic field is out of the page, as shown. The electric charge of each of the three particles is, respectively,



A) 1 is neutral, 2 is positive, and 3 is negative.
B) 1 is neutral, 2 is negative, and 3 is positive.
C) 1 is positive, 2 is neutral, and 3 is negative.
D) 1 is negative, 2 is neutral, and 3 is positive.
E) 1 is positive, 2 is negative, and 3 is neutral.

7) A proton is projected with a velocity of 7,000 m/s into a magnetic field of 0.60 T perpendicular to the motion of the proton. What is the force that acts on the proton?



$|\vec{v}| = 7 \times 10^{3}$ m/s
$|\vec{B}| = 0.6$ T
$\phi = 90°$
$q = 1.6 \times 10^{-19}$ C

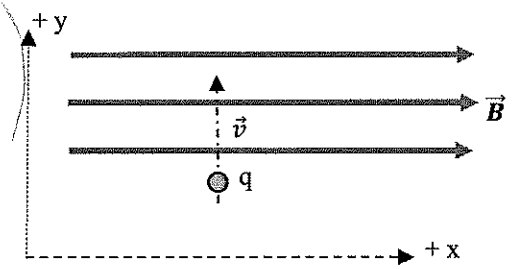
$|\vec{F}_m| = q|\vec{v}||\vec{B}|\sin\phi$
$|\vec{F}_m| = 6.72 \times 10^{-15}$ N

8) A stationary proton is in a uniform magnetic field of 0.20 T. What is the magnitude of the magnetic force on the proton?

$|\vec{v}| = 0$ m/s $\quad |\vec{F}_m| = 0$ N

9) A positively charged particle (q = 0.03 C) enters a magnetic field (B = 0.4 T) as shown with a velocity of 100 m/s, what is the magnitude and direction of the magnetic force acting on the particle?

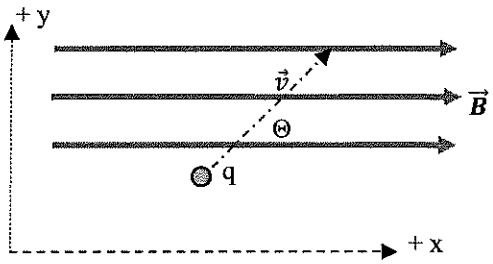
$|\vec{F}_m| = q|\vec{v}||\vec{B}|\sin\phi = 1.2$ N (into page)



10) A negatively charged particle (q = 0.03 C) enters a magnetic field (B = 0.4 T) as shown with a velocity of 100 m/s, if it enters at an angle of Θ = 30° to the magnetic field line, what is the magnitude and direction of the magnetic force acting on the particle?

$|\vec{F}_m| = q|\vec{v}||\vec{B}|\sin\phi$ $\quad \phi = 30°$

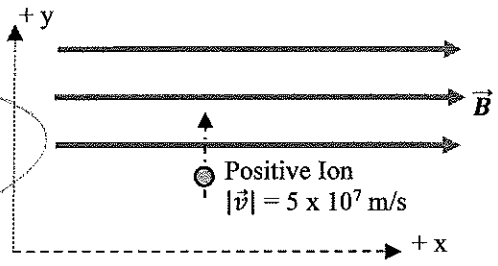
$|\vec{F}_m| = 0.6$ N into page



11) The uniform magnetic field $\vec{B}$ is along the plus x-axis and a positive ion is traveling with a velocity of v = 5 x 10^7 m/s in the plus y-direction perpendicular to the magnetic field. (Assume q = 1.6 x 10^{-19} C) If the magnitude of the force on the ion due to the magnetic field is 2.4 x 10^{-13} N, what is the magnitude of the Magnetic Field?

$|\vec{F}_m| = qvB\sin\phi$

$|\vec{B}| = \dfrac{|\vec{F}_m|}{q|\vec{v}|\sin\phi} = 0.03\,T$



12) A charged particle of mass 0.0020 kg is subjected to a 6 T magnetic field which acts at a right angle to its motion. If the particle moves in a circle of radius 0.2 m at a speed of 5 m/s what is the magnitude of the charge on the particle?

$\Sigma\vec{F} = \vec{F}_m = m\vec{a}_c$

$F_m = ma_c$

$qvB = \dfrac{mv^2}{r}$

Solve for q

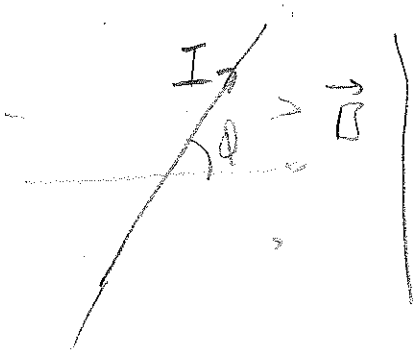
$q = \dfrac{mv^2}{rvB} = \dfrac{mv}{rB}$

$q = 0.0023\,C$

13) The magnetic force on a current-carrying wire in a magnetic field is the strongest when

A) the current is in the direction of the magnetic field lines.
B) the current is in the direction opposite to the magnetic field lines.
C) the current is at a 180° angle with respect to the magnetic field lines.
D) the current is perpendicular to the magnetic field lines.
E) the current is at a 0° angle with respect to the magnetic field lines.

14) A wire carrying a 2-A current is placed at an angle of 60° with the respect to a magnetic field of strength 0.2 T. If the length of the wire is 0.6 m what is the magnitude of the magnetic force acting on the wire?

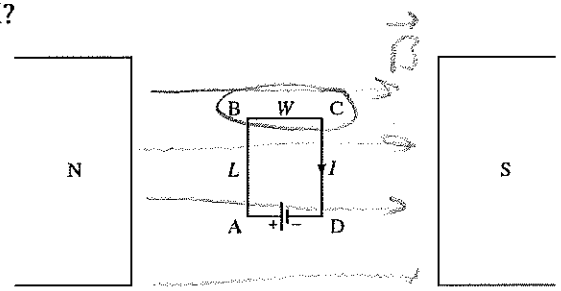


$I = 2A$
$\phi = 60°$
$L = 0.6m$
$|\vec{B}| = 0.2T$

$|\vec{F}_m| = IBL\sin\phi$
$|\vec{F}_m| = 0.2\,N$
into page

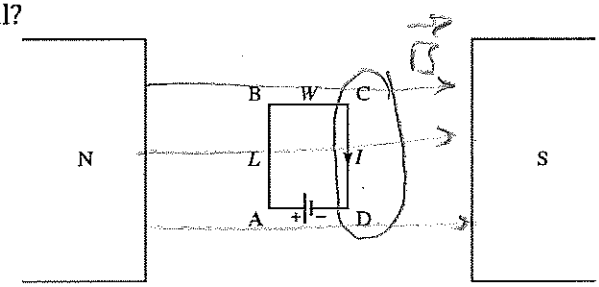
15) A rectangular coil, with corners labeled ABCD, has length *L* and width *w*. It is placed between the poles of a magnet, as shown in the figure If there is a current, *I* flowing through this coil in the direction shown, what is the direction of the force acting on section BC of this coil?

A) perpendicular to and into the page
B) perpendicular to and out of the page
C) in the direction of the magnetic field
D) in the opposite direction of the magnetic field
E) The force is zero.



16) A rectangular coil, with corners labeled ABCD, has length *L* and width *w*. It is placed between the poles of a magnet, as shown in the figure If there is a current, *I* flowing through this coil in the direction shown, what is the direction of the force acting on section CD of this coil?

A) perpendicular to and into the page
B) perpendicular to and out of the page
C) in the direction of the magnetic field
D) in the opposite direction of the magnetic field
E) The force is zero.



R11R #1

17) A mass spectrometer is used to determine the mass of a particle with a net positive charge of 2×10^{-6} C that is accelerated through a potential difference of 200 V into a magnetic field of 3 T. If the particle has a radius of 0.3 m as it moves through the field, what is the particles mass?

$m = \frac{q r^2 B^2}{2 \Delta V}$

$m = 4.05 \times 10^{-9}$ kg

$q = 2 \times 10^{-6}$ C
$r = 0.3$ m
$B = 3$ T
$\Delta V = 200$ V

18) A charged particle of mass 0.0020 kg is subjected to a 6 T magnetic field which acts at a right angle to its motion. An uniform electric field of 1200 N/C is also acting on the particle and is at right angles to the magnetic field, what velocity must the particle have to not be deflected as it moves through both fields?

$|\vec{v}| = \frac{|\vec{E}|}{|\vec{B}|} = \frac{1200 \text{ N/C}}{6 \text{ T}}$

200 m/s

KEY

1) B
2) B
3) C
4) B
5) D
6) D
7) 6.27×10^{-16} N
8) 0 N
9) 1.2 N into the page
10) 0.6 N out of the page
11) 0.03 T
12) 0.0083 C
13) D
14) 0.2 N into the page
15) E
16) B
17) 4.05×10^{-9} kg
18) 200 m/s